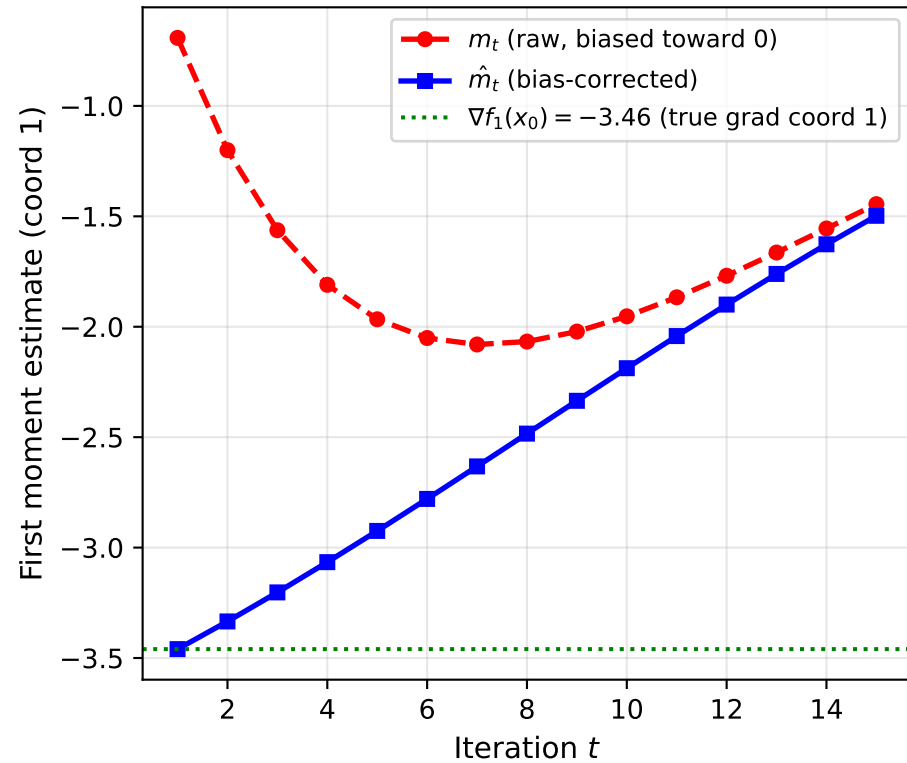


Adam Bias Correction: Impact on Moment Estimates and Step Sizes ($\beta_1 = 0.80$, $\beta_2 = 0.999$)

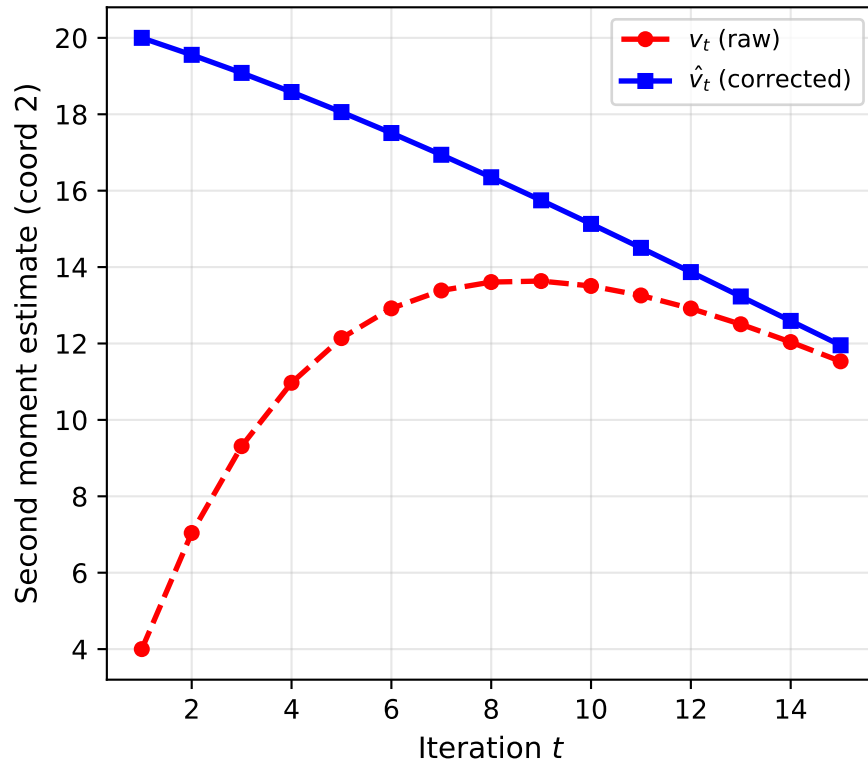
Bias Correction of First Moment

$$m_t \text{ vs } \hat{m}_t = m_t / (1 - \beta_1^t)$$



Bias Correction of Second Moment

$$v_t \text{ vs } \hat{v}_t = v_t / (1 - \beta_2^t)$$



Effective Step: Bias Correction Impact

(critical in early iterations)

